

1. A patient visits their doctor feeling unwell. The doctor thinks that there is a 0.7 probability that the patient has Statisticalitis. The main treatment for this (mild) illness is a week-long course of Prepazole. If the patient has Statisticalitis then the doctor believes that Prepazole will enable them to make a complete recovery with probability 0.9. If the patient does not have Statisticalitis, then the probability that the patient makes a full recovery after a course of Prepazole is only 0.6.
 - a. What is it about the way the information is presented in this question that suggests that you might usefully represent it using a tree diagram?
 - b. Represent this information on a tree diagram, making sure to clearly define any events that you use.
 - c. Find the probability that the patient makes a full recovery after a week-long course of Prepazole.
 - d. Find the probability that the patient had Statisticalitis, given that they make a full recovery after a week-long course of Prepazole.
 - e. Find the probability that the patient did not have Statisticalitis, given that they did not make a full recovery after a week-long course of Prepazole.
2. Pete thinks that the probability that a team from Manchester will win this year's Premier League is 0.3. If the winning team is from Manchester, then he thinks there is a 0.8 probability that the team will play in a mostly blue kit, and a 0.2 probability that the team will play in a mostly red kit. If the winning team is not from Manchester, then he thinks there is a 0.1 probability that the team will play in a mostly blue kit and a 0.5 probability that the team will play in a mostly red kit.
 - a. Find a way to represent this on a tree diagram. Ensure that your tree diagram is complete, even if the information above is not. Clearly define the events you are using.
 - b. Find the probability that the winning team is from Manchester, given that the winning team plays in a mostly red kit.
3. In the final of a game show contestants are asked two questions. The producers of the show model the chance that a contestant will get the first question right with a probability of 0.6. If the contestant gets the first question right, then they are asked a hard question for the star prize; otherwise, they win a consolation prize. Model the chance that such a contestant gets the hard question correct with a probability of 0.1. If the contestant gets the first question incorrect, then they are asked an easier question; if they get this right, they are given the consolation prize. Model the chance that such a contestant gets the easy question correct with a probability of 0.5. According to this model, find the probability that a contestant:
 - a. wins the star prize
 - b. wins the consolation prize
 - c. gets the first question correct, given that they win the consolation prize.
4. Jaylen and Elwood are both taking a French vocab test. Their friend Oliver thinks the probability that Jaylen will pass is 0.7 and that the probability that Elwood will pass is 0.6. He also thinks there is a 0.5 probability that they will both pass.
 - a. Given that Elwood passes the test, find the probability that Jaylen passes the test, according to Oliver's model.
 - b. Explain how you know that Oliver's model does not treat the events 'Elwood passes' and 'Jaylen passes' as independent. How can this make sense?
5. Two events A and B are such that $P(A|B) = 0.6$, $P(B|A) = 0.3$, and $P(A \cup B) = 0.9$. Find $P(A)$.

6. Suzanna and Yusef each attempt to throw a pen into a bin from a distance. Roan models the chance that Suzanna gets the pen in with a probability of 0.7 and the chance that Yusef gets the pen in with a probability of 0.6. He thinks there is a probability of 0.1 that both will miss.
 - a. Show that the event ‘Suzanna gets the pen in the bin’ and the event ‘Yusef gets the pen in the bin’ are not independent. How can this make sense?
 - b. Given that at least one person successfully throws their pen into the bin, find the probability that exactly one person successfully throws their pen into the bin.
7. A motorist's journey home involves passing through three sets of traffic lights. The motorist models the chances of being delayed at each set of lights as independent, and with probabilities 0.2, 0.6, and 0.5. Find, according to this model, the probability that the motorist:
 - a. is delayed at all three sets of lights
 - b. is delayed at exactly one set of lights
 - c. is delayed at the second set of lights, given that she is delayed at exactly one set of lights.
8. Two events G and H are such that $P(G) = 0.4$, $P(H) = 0.25$ and $P(G \cup H) = 0.55$. Show that G and H are independent.
9. Michelle is practicing free-throws in basketball. Her coach models the chance that she scores a basket with her first attempt as 0.85. If she scores, then the probability that she scores with her next attempt is 0.95. If she doesn't score, then the probability that she scores with her next attempt is 0.75. If Michelle repeatedly attempts this shot, then what is the probability that:
 - a. she scores on each of her first ten attempts
 - b. her first miss is on her fifth attempt
 - c. her first success is on her third attempt
 - d. she misses exactly once in the first five attempts.
10. Manpreet is practicing ten-pin bowling. He models the chance that he gets a strike in his very first frame as 0.4. Thereafter, if he gets a strike in his previous frame then there is a 0.8 probability that he gets a strike in the current frame. If he did not score a strike in his previous frame then there is a 0.6 probability that he gets a strike in the current frame. What is the probability that Manpreet:
 - a. bowls a perfect game (12 strikes in a row);
 - b. gets his first strike in frame five
 - c. gets a strike in frame four given that he gets a strike in frame five.

Without doing further calculations, guess the probability that Manpreet gets a strike in Frame ten? Can you find a way to justify this using only a small amount of calculation?

If S_n is the event ‘Manpreet gets a strike in frame n ’ then does what you wrote for the above mean that S_n and S_{n+1} are (approximately) independent for sufficiently large n ?